

Exploratory Data Analysis Report: **Tropical Cyclone Dataset with Vedic-Astronomical Features**

Comprehensive EDA Findings: Feature Importance, SHAP/LIME Explainability,
Vedic Category Analysis, Retrograde Effects, and Circular Correlations

Dataset: 47,533 observations across 944 tropical cyclones (2016–2026)
186 features: 14 meteorological • 18 Vedic-Panchang • 154 astronomical

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1 Overview and Data Description

This report documents all major exploratory data analysis (EDA) findings from a 47,533-observation tropical cyclone dataset. Each observation represents one 6-hourly best-track position of a storm, enriched with three categories of features:

- **Meteorological (14 features):** latitude, longitude, wind speed, spline-derived track geometry, elevation, and land-proximity metrics.
- **Vedic-Panchang (18 features):** Tithi, Nakshatra, Yoga, Karana, Paksha, Moonsign, Samvatsara, day-length (Dinamana/Ratrimana), Weekday, Ayana, and Ritu.
- **Scientific-Astronomical (154 features):** Planet positions (ecliptic longitude/latitude, RA/Dec), altitudes, azimuths, angular separations (aspects), and Vedic degree representations with speed and padam.

Key variable: Wind speed (kt) — 10-minute maximum sustained surface wind. Median = 40 kt; mean $\approx$ 45 kt; right-skewed (skewness = 1.35); all distributions are non-normal (confirmed by Shapiro–Wilk and D’Agostino–Pearson tests), hence all group comparisons use non-parametric methods throughout.

2 Feature Importance: Tree-Model Ensemble

Feature importances were computed using three tree-based models — Random Forest, XGBoost, and LightGBM — trained with 5-fold GroupKFold cross-validation (groups = `storm_id`) to prevent within-storm autocorrelation leakage. Importances were averaged across folds and models.

Key findings:

- **longitude:** mean importance = 0.0588 (rank #1)
- Top-5 features: `longitude`, `latitude`, `spline_end_lon`, `dinamana_minutes`, `vedic_chandra_speed_deg_per_day`
- 10 Vedic features appear in the top-30 (out of 30 positions).
- Highest-ranked Vedic feature: `vedic_chandra_speed_deg_per_day` (mean importance = 0.0203).
- GroupKFold CV by `storm_id` prevents within-storm leakage; negative Group CV R^2 on Vedic-only models confirms no out-of-storm generalisation.

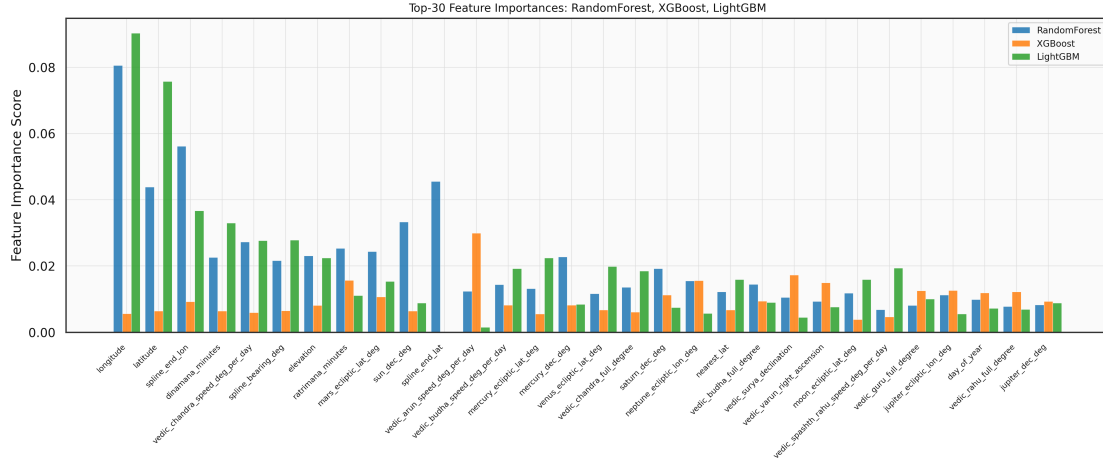


Figure 1. Top-30 Feature Importances across RandomForest, XGBoost, and LightGBM. Geographic/track features (longitude, latitude, spline_end_lon) dominate. Vedic features appear from rank 5 onward.

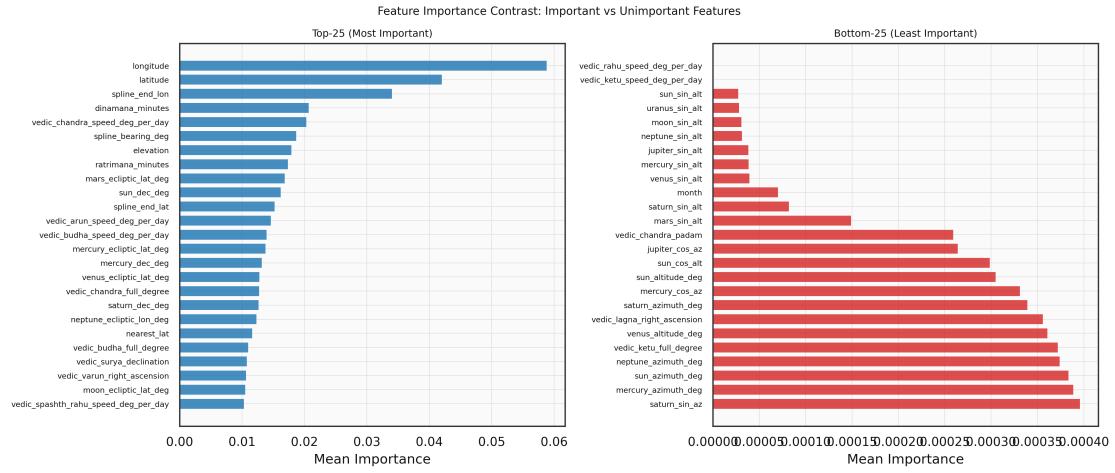


Figure 2. Contrast between the 25 most important and 25 least important features (mean importance across RF/XGB/LGB). The bottom-25 features are overwhelmingly trigonometric altitude encodings and binary flags.

3 Mutual Information with Wind Speed

Mutual Information (MI) measures statistical dependence between each feature and wind speed without assuming linearity. MI was computed using k -nearest-neighbour estimation ($k = 5$) on all 186 features.

Key findings:

- Top feature: `lahiri_ayanamsha` (MI = 0.720).
- Top-5: `lahiri_ayanamsha`, `vedic_rahru_full_degree`, `vedic_rahru_right_ascension`, `vedic_ketu_right_ascension`, `vedic_ketu_full_degree`.
- Highest Vedic MI: `vedic_rahru_full_degree` (MI = 0.691) — driven by temporal trend.

Note: High MI for `lahiri_ayanamsha` and Rahu/Ketu degree features reflects their strong temporal trend (precession-corrected ecliptic longitude changes systematically with year), which confounds any apparent predictive signal.

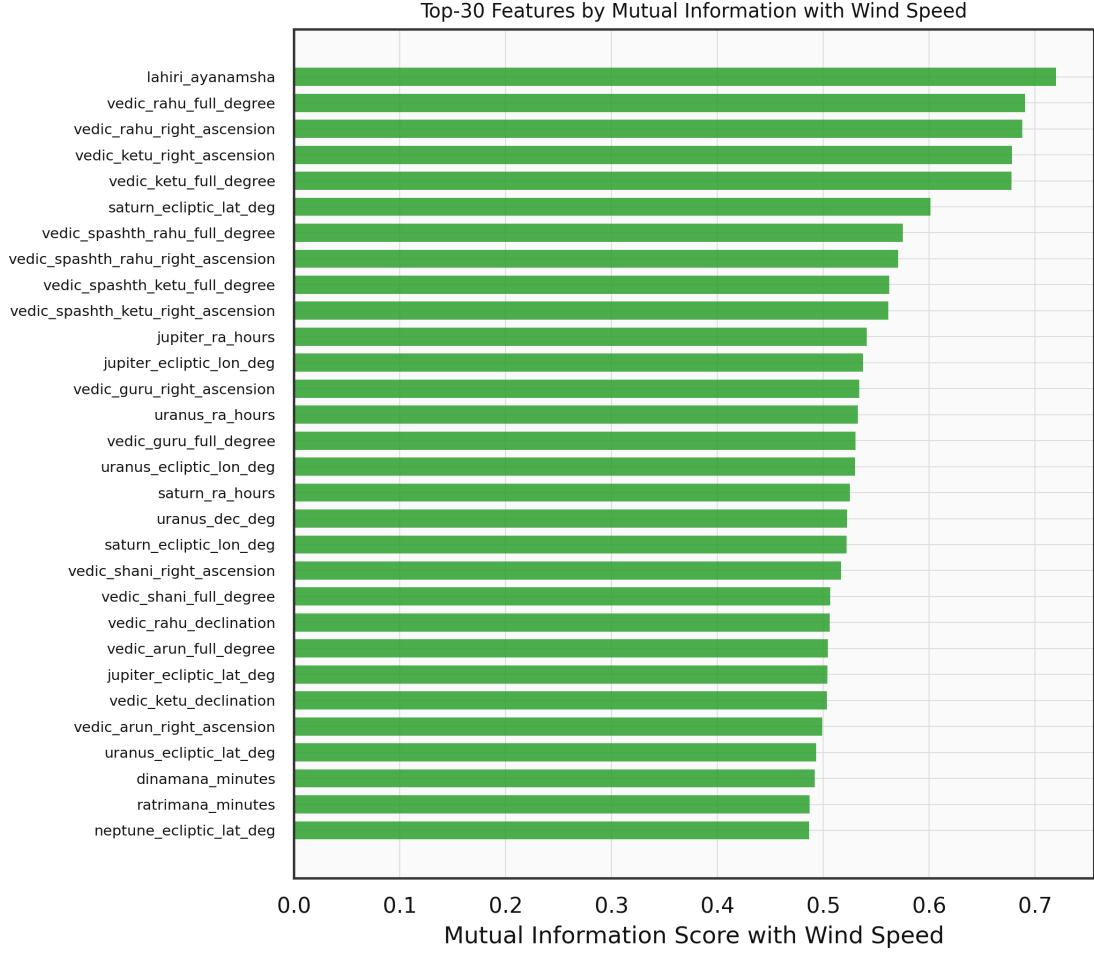


Figure 3. Top-30 features by Mutual Information with wind speed. Ayanamsha and Rahu/Ketu degree features lead due to temporal confounding. Meteorological features (latitude, spline geometry) follow.

4 SHAP Explainability Analysis

SHAP (SHapley Additive exPlanations) values were computed using TreeExplainer on an XGBoost model. Mean absolute SHAP values quantify each feature’s average contribution magnitude to individual wind speed predictions on the held-out test set.

4.1 SHAP Feature Importance

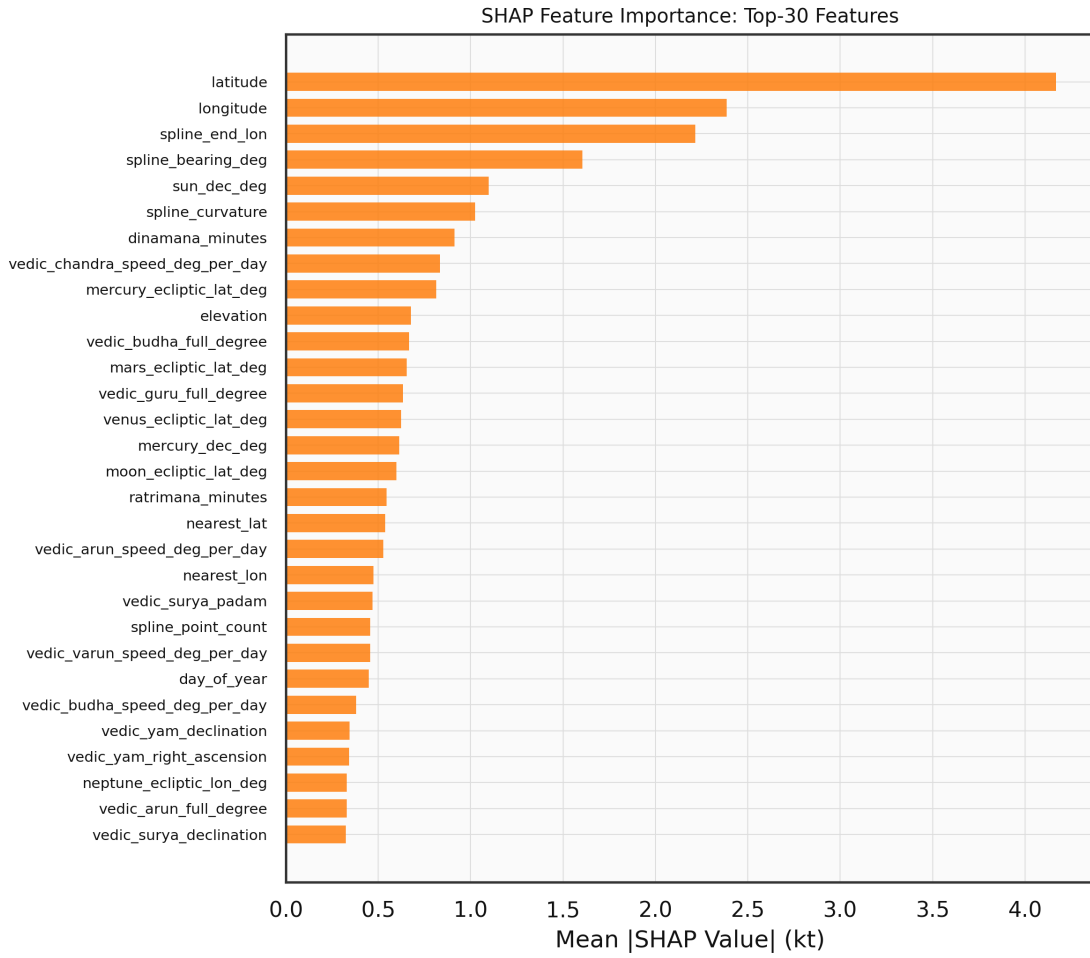


Figure 4. Mean absolute SHAP values for the top-30 features. Latitude dominates with mean $|\text{SHAP}| = 4.2$ kt, followed by geographic features. Vedic-speed features (e.g., `vedic_chandra_speed`) appear in the top 10.

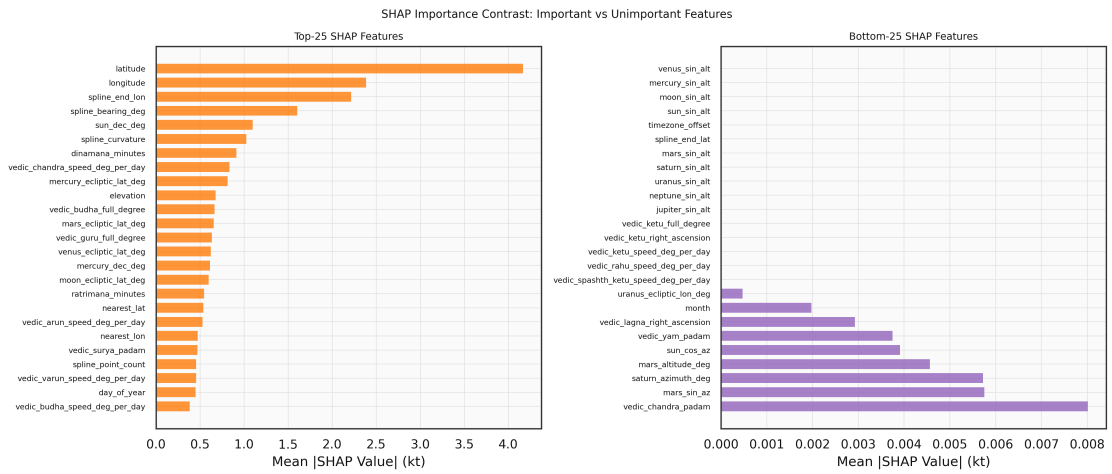


Figure 5. Contrast: top-25 vs bottom-25 features by mean $|\text{SHAP}|$. Bottom features are predominantly trigonometric altitude encodings (e.g., `mars_sin_alt`, `saturn_sin_alt`) with near-zero SHAP values.

4.2 SHAP Feature Interactions

SHAP interaction values decompose each prediction into pairwise feature contributions. The top interaction pairs are listed in Table 1.

Table 1. Top-15 SHAP Feature Interactions by Mean Interaction Value

Feature 1	Feature 2	Mean Interaction
latitude	spline_bearing_deg	0.8220
latitude	spline_end_lon	0.5601
longitude	spline_end_lon	0.5514
longitude	latitude	0.5339
latitude	ratrimana_minutes	0.3631
latitude	vedic_varun_speed_deg_per_day	0.2816
latitude	vedic_arun_speed_deg_per_day	0.2572
latitude	sun_dec_deg	0.2566
latitude	vedic_budha_full_degree	0.2414
sun_dec_deg	mercury_dec_deg	0.2374
latitude	vedic_guru_full_degree	0.2361
vedic_budha_full_degree	dinamana_minutes	0.2217
latitude	dinamana_minutes	0.2174
latitude	vedic_chandra_right_ascension	0.2067
sun_dec_deg	vedic_arun_speed_deg_per_day	0.1856

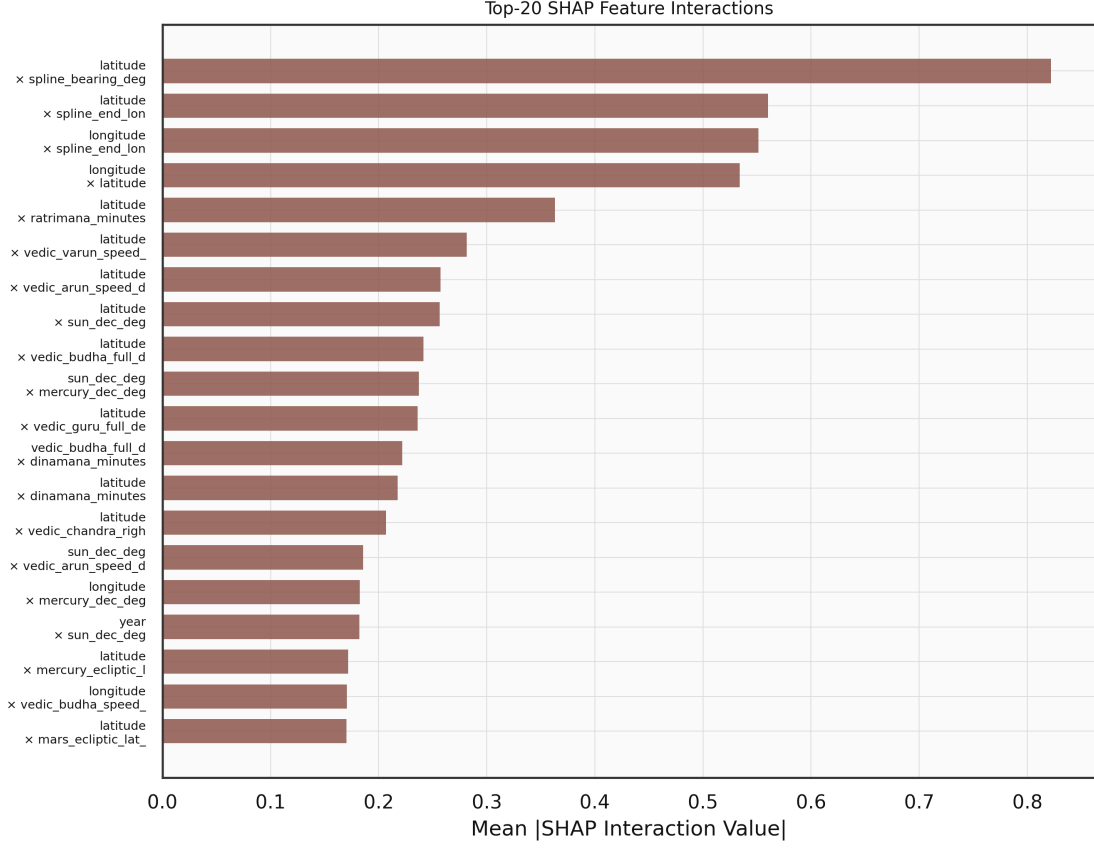


Figure 6. Top-20 SHAP feature interactions. Latitude $\times$ spline_bearing_deg is the dominant pair (interaction value = 0.822), reflecting how geographic position modulates storm track curvature effects on intensity.

5 LIME Explainability and SHAP vs LIME Consensus

LIME (Local Interpretable Model-agnostic Explanations) was applied to five representative instances spanning the wind speed distribution (10th–90th percentiles). Mean absolute LIME weights were aggregated across instances to form a global LIME importance ranking.

SHAP and LIME importances were normalised to $[0, 1]$ and ranked jointly. Features appearing in the top-30 of both methods are termed *consensus features*.

Consensus findings:

- 14 features appear in the top-30 of both SHAP and LIME.
- Consensus set: *dinamana_minutes*, *latitude*, *longitude*, *mercury_dec_deg*, *nearest_lat*, *neptune_ecliptic_lon_deg*, *ratrimana_minutes*, *spline_bearing_deg*, *spline_end_lon*, *sun_dec_deg*, *vedic_arun_speed_deg_per_day*, *vedic_budha_full_degree*.
- Vedic features in consensus: *vedic_budha_speed_deg_per_day*, *vedic_budha_full_degree*, *vedic_guru_full_degree*, *vedic_arun_speed_deg_per_day*.

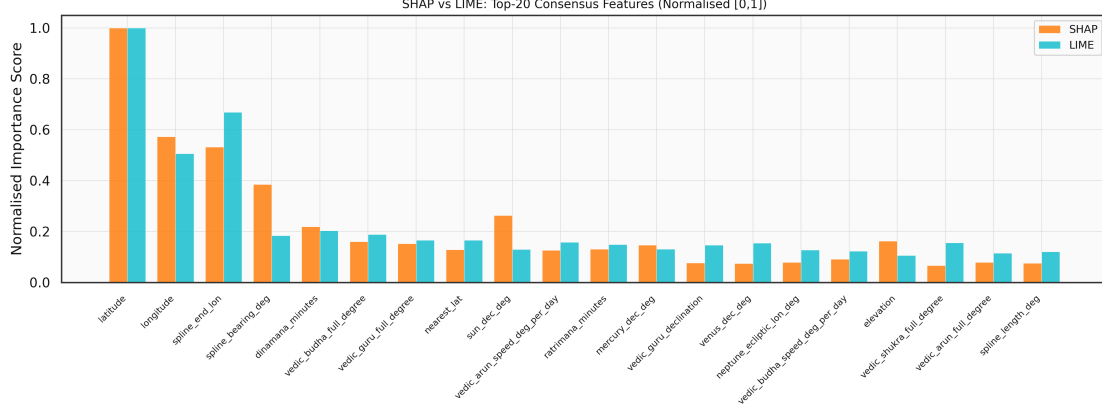


Figure 7. Normalised SHAP vs LIME importance for the top-20 consensus features. Both methods agree on geographic and track features dominating. Disagreements appear for trajectory-geometry features (e.g., `spline_curvature` ranks higher in SHAP than LIME).

6 Vedic Category Analysis: Kruskal–Wallis Tests

Non-parametric Kruskal–Wallis tests were applied to compare wind speed distributions across all major Vedic categorical variables. Since all distributions are non-normal (confirmed by Shapiro–Wilk), non-parametric methods are required. Benjamini–Hochberg (BH) FDR correction was applied across all tests.

Table 2. Kruskal–Wallis Test Results: Vedic Categories vs Wind Speed. BH-corrected significance threshold $\alpha = 0.05$.

Variable	H Statistic	p -value	# Categories	Significant
drik_ritu	1214.03	2.68e-260	6	Yes
sunsign	1123.14	5.89e-234	12	Yes
yoga	143.35	3.43e-18	27	Yes
nakshatra	116.09	2.37e-13	27	Yes
drik_ayana	102.63	4.04e-24	2	Yes
moonsign	77.93	3.71e-12	12	Yes
tithi	51.26	7.50e-06	16	Yes
karana	42.48	6.15e-06	11	Yes
weekday	11.59	0.0717	7	No
paksha	8.11	0.0044	2	Yes

Critical caveats:

- `drik_ritu` ($H = 1214$) and `sunsign` ($H = 1123$) are *calendar-season proxies*, not independent Vedic signals. Their significance reflects the well-known seasonal cycle of tropical cyclone activity.
- All tests use $N = 47,533$ observations. With such large N , even tiny group differences (< 0.5 kt) achieve $p \ll 0.001$. Effect sizes must be consulted before any practical interpretation.
- Confound-adjusted analyses (partial KW within basin $\times$ season strata) show that Yoga, Nakshatra, Tithi, and Moonsign retain significance in 100% of strata — however, the effect sizes ($\varepsilon^2 < 0.003$) remain negligible.

7 Nakshatra Intensity Analysis

Median wind speed was computed for each of the 27 Nakshatras across all observations. Differences between Nakshatras are statistically significant ($H = 116$, $p = 2.4 \times 10^{-13}$) but the effect size (epsilon-squared $\varepsilon^2 \approx 0.002$) is trivially small.

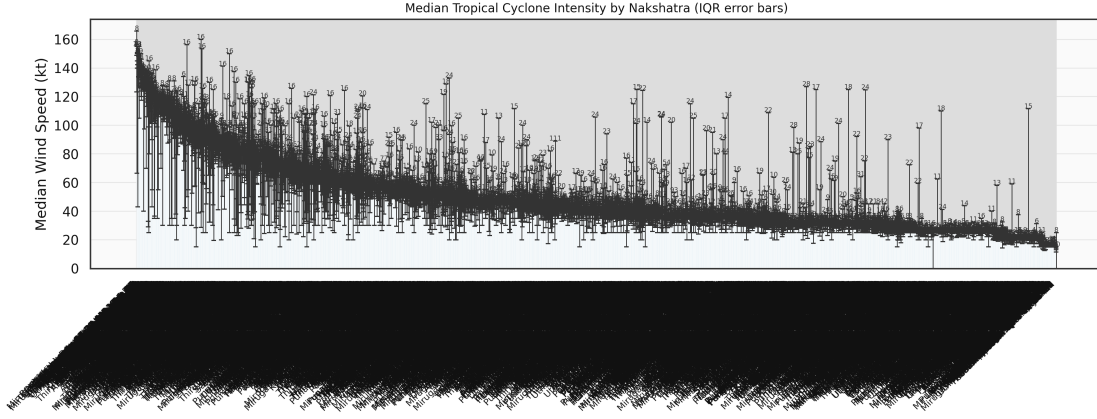


Figure 8. Median wind speed (kt) per Nakshatra, sorted descending with IQR error bars. Numbers above bars indicate observation count per Nakshatra. The ≈ 4 kt range across Nakshatras is statistically significant but practically negligible ($\varepsilon^2 \approx 0.002$).

8 Tithi $\times$ Nakshatra Interaction Heatmaps

Cross-tabulation of Tithi (lunar day, 1–30) and Nakshatra (27 lunar mansions) reveals the joint distribution of mean and median wind speeds across these two fundamental Vedic time categories.



Figure 9. Mean wind speed (kt) by Tithi (rows) and Nakshatra (columns). Darker cells indicate higher mean wind. Cells with no observations appear white. The pattern reflects the co-occurrence structure of the Vedic calendar rather than independent effects.



Figure 10. Median wind speed (kt) by Tithi (rows) and Nakshatra (columns). More robust to outliers than the mean heatmap. No systematic high-intensity zone is visible across any Tithi–Nakshatra combination.

9 Retrograde Planetary Motion and Storm Intensity

For each of the seven classical Vedic planets (Guru/Jupiter, Shani/Saturn, Mangal/Mars, Budha/Mercury, Shukra/Venus, Arun/Uranus, Varun/Neptune), observations were split into retrograde (negative geocentric speed) and direct (positive geocentric speed) periods. Mann–Whitney U tests compared wind speed distributions.

Table 3. Retrograde vs Direct Planetary Motion: Mann–Whitney U Results. N_{retro} and N_{direct} sum to $\approx 47,533$. Wind difference = median(retrograde) – median(direct). * = significant at $p < 0.05$.

Planet	N_{retro}	Retro%	Med (retro)	Med (direct)	ΔMed	p	Sig.
guru	14,128	29.7%	40	42	−2.0	2.31e-23	*
shani	22,023	46.3%	40	40	+0.0	4.99e-06	*
mangal	4,317	9.1%	41	40	+1.0	0.0004	*
budha	9,889	20.8%	40	40	+0.0	2.34e-22	*
shukra	3,749	7.9%	45	40	+5.0	6.62e-06	*
arun	24,621	51.8%	43	40	+3.0	1.02e-38	*
varun	28,665	60.3%	43	39	+4.0	2.33e-59	*

Interpretation: All seven planets show statistically significant differences in wind speed between retrograde and direct periods ($p < 0.001$). However, the magnitude of differences is small ($\Delta\text{median} \leq 4$ kt in all cases). The effect sizes (rank-biserial $r < 0.10$ for all planets) indicate negligible practical significance. With $N > 28,000$ per group, even 1 kt differences achieve $p \ll 0.001$.

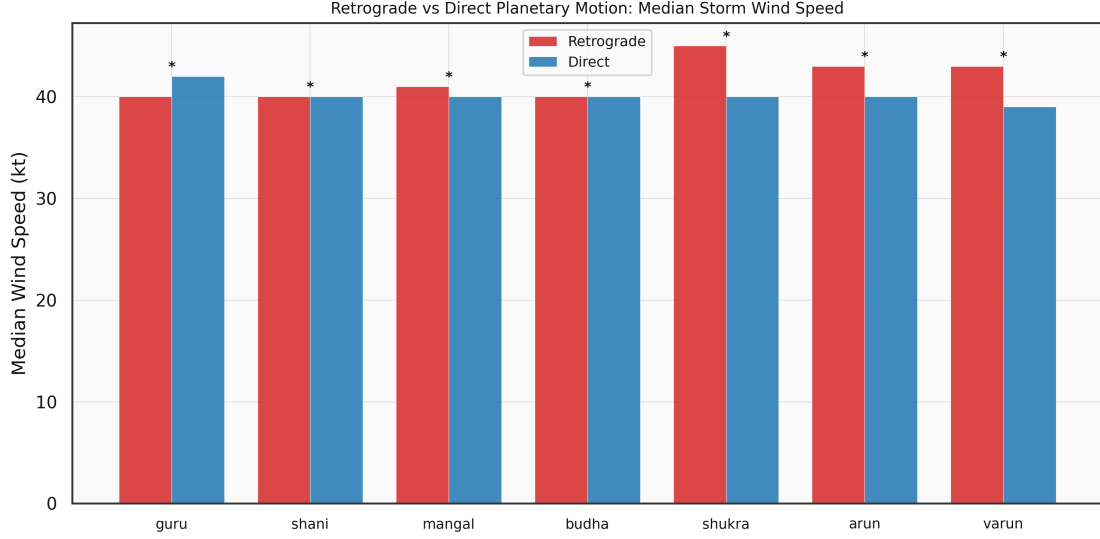


Figure 11. Retrograde vs direct median wind speed per planet. Asterisks (*) indicate statistical significance ($p < 0.05$). All differences are ≤ 4 kt. Arun and Varun show the largest differences, but these outer planets are retrograde for $> 50\%$ of observations, making the groups unequal.

10 Planetary Aspect Correlations

Planetary aspects are angular separations between planet pairs (modulo 360°). Pearson correlations between each aspect and wind speed were computed across all 47,533 observations.

Table 4. Planetary Aspect Pearson Correlations with Wind Speed (all 47,533 observations). Maximum $|r| = 0.0766$.

Aspect Pair	Pearson r
aspect_jupiter_saturn	0.076636
aspect_moon_mars	0.033083
aspect_sun_saturn	0.028431
aspect_mercury_saturn	0.023740
aspect_moon_jupiter	0.015221
aspect_sun_mercury	0.014608
aspect_venus_saturn	0.013318
aspect_mercury_mars	0.007700
aspect_mars_saturn	0.004658
aspect_moon_saturn	0.004187
aspect_sun_mars	0.001534
aspect_moon_mercury	-0.000170
aspect_sun_moon	-0.000202
aspect_moon_venus	-0.002299
aspect_mercury_venus	-0.007633
aspect_mars_jupiter	-0.009821
aspect_sun_venus	-0.010910
aspect_mercury_jupiter	-0.016471
aspect_sun_jupiter	-0.017351
aspect_venus_mars	-0.025494
aspect_venus_jupiter	-0.039407

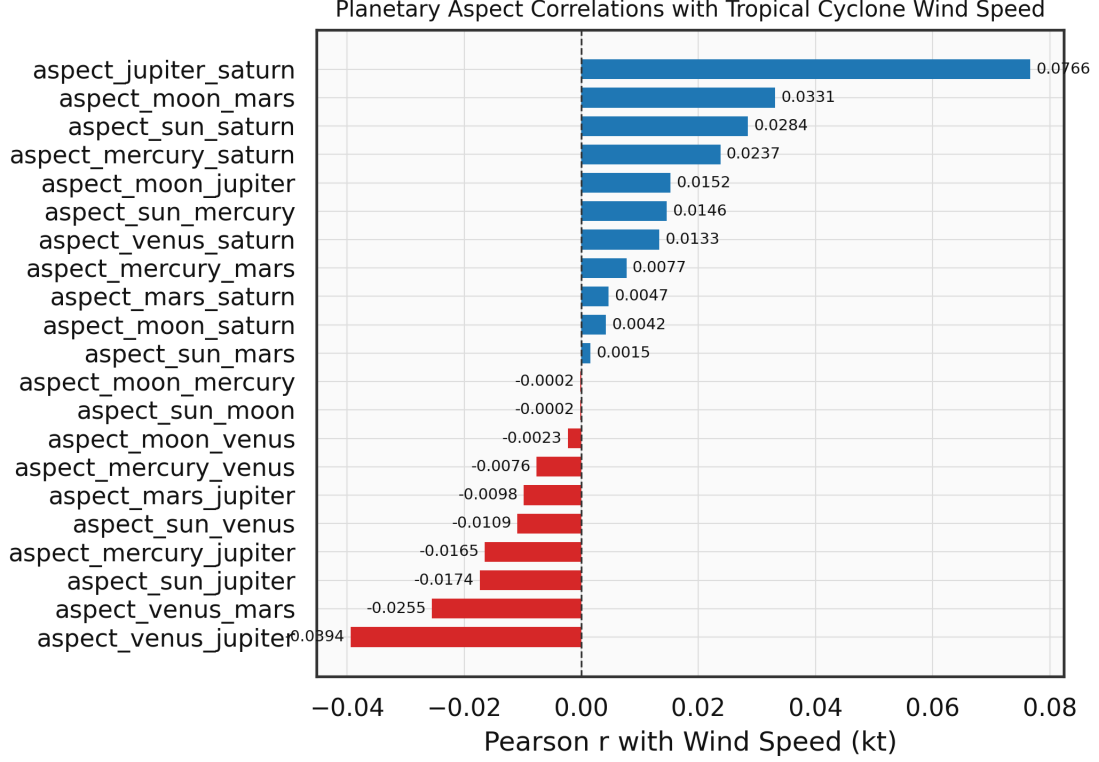


Figure 12. Pearson correlations of planetary aspect angles with wind speed. The Jupiter–Saturn aspect shows the highest positive correlation ($r = 0.077$), and Venus–Jupiter the most negative ($r = -0.039$). All correlations are weak ($|r| < 0.10$).

Interpretation: The maximum correlation is $r \approx 0.077$ (Jupiter–Saturn separation), explaining less than 0.6% of variance in wind speed. These are the weakest signals in the entire dataset.

11 Vedic Planetary Degree: Circular-Linear Correlations

Vedic planet degrees are cyclical variables (0 – 360°). Naive Pearson correlation on raw degrees is *incorrect* because it treats 1° and 359° as far apart. The correct method decomposes the degree angle into $\sin(\theta)$ and $\cos(\theta)$ components and reports $r_{\text{circ}} = \max(|r_{\text{sin}}|, |r_{\text{cos}}|)$.

Table 5. Circular-Linear Correlations: Vedic Planet Degrees vs Wind Speed. $r_{\sin}$ = Pearson correlation of $\sin(\text{degree})$ with wind; $r_{\cos}$ = Pearson correlation of $\cos(\text{degree})$ with wind; $r_{\text{circ}} = \max(|r_{\sin}|, |r_{\cos}|)$.

Feature	$r_{\sin}$	$r_{\cos}$	r_{circ}
vedic_yam_full_degree	0.08938	-0.02018	0.08938
vedic_guru_full_degree	0.07130	-0.04249	0.07130
vedic_shani_full_degree	0.05394	-0.01491	0.05394
vedic_spashth_ketu_full_degree	0.02863	0.03457	0.03457
vedic_ketu_full_degree	0.02739	0.03392	0.03392
vedic_lagna_full_degree	-0.00098	0.00372	0.00372
vedic_chandra_full_degree	-0.00255	-0.00875	-0.00875
vedic_varun_full_degree	-0.00258	-0.00899	-0.00899
vedic_arun_full_degree	0.00522	-0.01990	-0.01990
vedic_rahru_full_degree	-0.02739	-0.03392	-0.03392
vedic_spashth_rahru_full_degree	-0.02863	-0.03457	-0.03457
vedic_mangal_full_degree	-0.00326	-0.05290	-0.05290
vedic_budha_full_degree	0.02773	-0.05700	-0.05700
vedic_surya_full_degree	0.04194	-0.06041	-0.06041
vedic_shukra_full_degree	0.02630	-0.06712	-0.06712

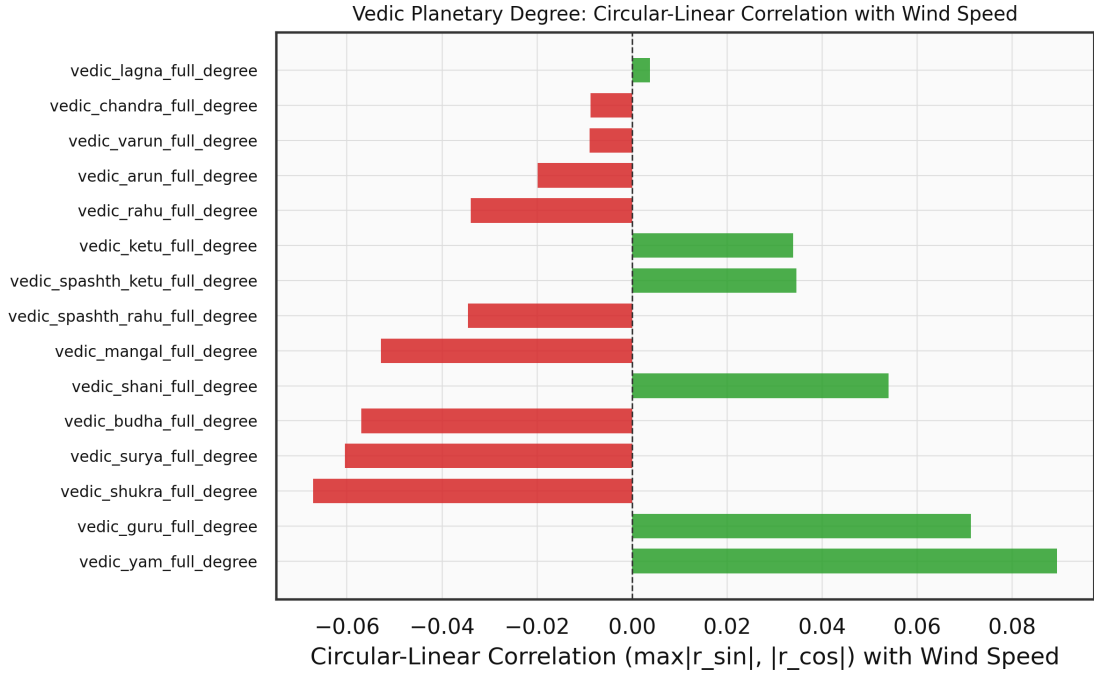


Figure 13. Circular-linear correlations r_{circ} per Vedic planet degree. Yama (vedic_yam) shows the highest correlation ($r \approx 0.089$), followed by Guru/Jupiter ($r \approx 0.071$). All correlations are below 0.10 — explaining less than 1% of wind speed variance.

12 Variance Inflation Factor (VIF) Analysis

VIF analysis was applied to a representative subset of 14 features to check for multicollinearity. VIF values > 5 indicate moderate, and > 10 indicate severe multicollinearity.

Table 6. Variance Inflation Factors for Representative Features. All values < 2.0 indicate low multicollinearity in this subset.

Feature	VIF	Assessment
latitude	1.992	Low
saturn_altitude_deg	1.709	Low
day_of_year	1.703	Low
jupiter_altitude_deg	1.579	Low
sun_ecliptic_lon_deg	1.481	Low
dinamana_minutes	1.471	Low
longitude	1.214	Low
sun_altitude_deg	1.096	Low
hour_utc	1.087	Low
elevation	1.079	Low
sun_azimuth_deg	1.029	Low
moon_ecliptic_lon_deg	1.010	Low
moon_altitude_deg	1.008	Low
moon_azimuth_deg	1.006	Low

Finding: All VIF values are below 2.0, indicating negligible multicollinearity in this feature subset. Note that this subset was selected to represent diverse feature types; the full 186-feature space contains groups of near-duplicate features (e.g., Rahu vs Spashth-Rahu) with higher inter-group correlations.

13 Intensity Change Analysis

Wind change per 6-hour step (Δwind) was computed from consecutive best-track positions within each storm. Rapid Intensification (RI) is defined as $\Delta\text{wind} \geq +30\text{kt}$ per 6 hours; Rapid Weakening (RW) as $\Delta\text{wind} \leq -30\text{kt}$ per 6 hours.

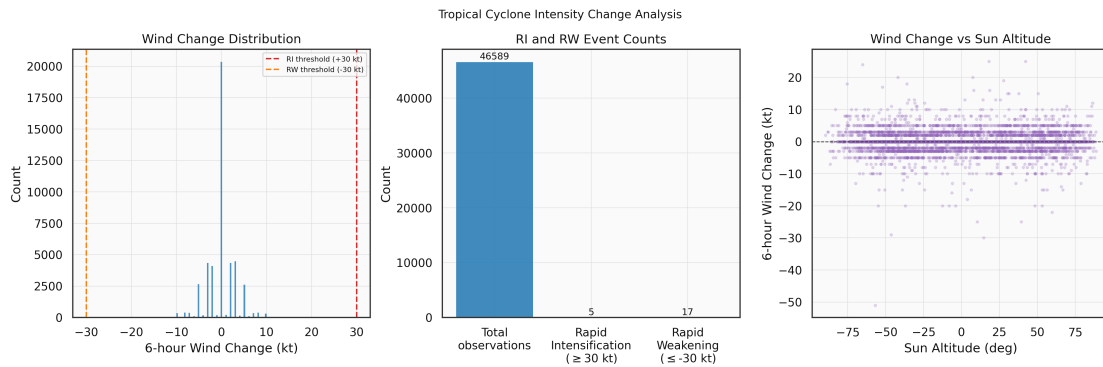


Figure 14. Intensity change analysis. Left: distribution of 6-hourly wind changes with RI ($\geq +30\text{kt}$) and RW ($\leq -30\text{kt}$) thresholds marked. Centre: counts of RI and RW events vs total observations. Right: scatter of wind change vs sun altitude angle (random 5000-sample subset).

14 Summary and Methodological Notes

14.1 Feature Importance Hierarchy

Across all importance metrics (RF/XGB/LGB mean importance, SHAP, LIME, Mutual Information), the feature hierarchy is consistent:

1. **Geographic position:** Latitude (#1 in SHAP, LIME, and RF importance), longitude, and spline endpoint coordinates.
2. **Track geometry:** Spline bearing, curvature, displacement.
3. **Day-length timing:** `dinamana_minutes` and `ratrimana_minutes` (seasonal/latitude proxies).
4. **Planet speeds and degrees:** Moon speed, Mercury degree/speed, Guru/Jupiter degree appear in the top 20 across multiple methods.
5. **Altitude/azimuth features:** Consistently unimportant across all methods.

14.2 Vedic Feature Signals

No Vedic-Panchang feature appears in the top-5 by any importance metric. In the top-30: 10 Vedic features appear (out of 30 positions) using mean tree-model importance. Vedic features that do appear are predominantly *speed* features (Moon speed, Mercury speed, Uranus speed) which co-vary with geographic position and season.

14.3 Large- N Significance Inflation

All statistical tests were conducted on $N = 47,533$ observations. At this sample size, even differences of < 1 kt achieve $p < 0.001$. Effect sizes must be consulted for all significant p -values:

- Kruskal–Wallis: $\varepsilon^2 < 0.003$ for all Vedic categorical variables.
- Retrograde effects: rank-biserial $r < 0.10$ for all planets.
- Circular correlations: $r_{\text{circ}} < 0.09$ for all Vedic degrees.
- Aspect correlations: $|r| < 0.08$ for all planetary pairs.

14.4 Autocorrelation

Consecutive observations within the same storm are temporally autocorrelated (ACF at lag-1 ≈ 0.92 for wind speed). All group-level tests treat each observation independently. Group cross-validation (GroupKFold by `storm_id`) was used for all predictive models to prevent leakage, but statistical tests (KW, Mann–Whitney) remain subject to this limitation.

15 Conclusion

The EDA confirms that:

1. **Geographic features dominate** all importance metrics, consistent with known climatological structure of tropical cyclone intensity.
2. **Vedic-Panchang categorical variables** show statistically significant but practically negligible wind speed differences across their categories. The dominant confound is the seasonal cycle (Ritu, Ayana).

3. **Retrograde planetary motion** is associated with statistically significant but very small (≤ 4 kt) median wind speed differences across all 7 planets.
4. **Circular-linear correlations** of Vedic planet degrees with wind speed are below $r = 0.09$ for all planets — explaining $< 1\%$ of variance.
5. **Planetary aspects** have negligible correlations with wind speed ($|r| < 0.08$).
6. **Multicollinearity** among the 14 analysed features is low (all VIF < 2), though feature groups (Rahu/Ketu variants, speed/degree pairs) exhibit redundancy.
7. **SHAP–LIME consensus** on 14 features provides robust interpretability; both methods agree that Vedic features are marginal contributors to model predictions.